Machine learning for electricity price forecasting

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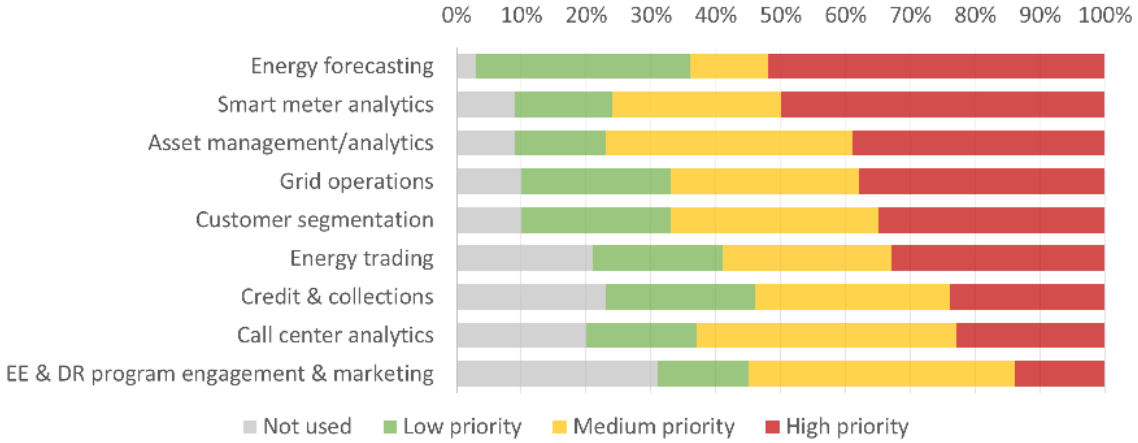


* Based on joint work with K.Bilińska, Y.Chawla, K.Chęć, A.Lipiecki, K.Maciejowska, W.Nitka, T.Serafin, B.Uniejewski, P.Zaleski (Wrocław Tech), C.Challu, K.Olivares (Nixtla), T.Hong (UNCC), K.Hubicka (UBS), A.Jędrzejewski (U.Aveiro), C.Kath (RWE), N.Kourentzes (U.Skövde), J.Lago (Amazon), G.Marcjasz (Alpiq), M.Narajewski (Statkraft), J.Nasiadka (Nokia), J.Nowotarski (BNY Mellon), H.Zareipour (U.Calgary), F.Ziel (U.Duisburg-Essen)

** Julia/Python notebooks coded by A.Lipiecki

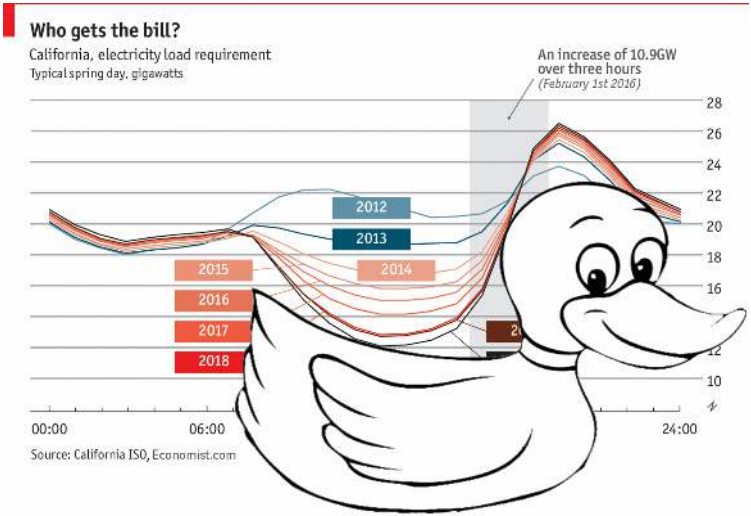
Energy forecasting is a top priority for utilities

(SAS, 2017, White Paper; responses from 136 utilities from 24 countries)




THE WORLD IS CHANGING ...

... increased (wind and) solar penetration take a toll ...

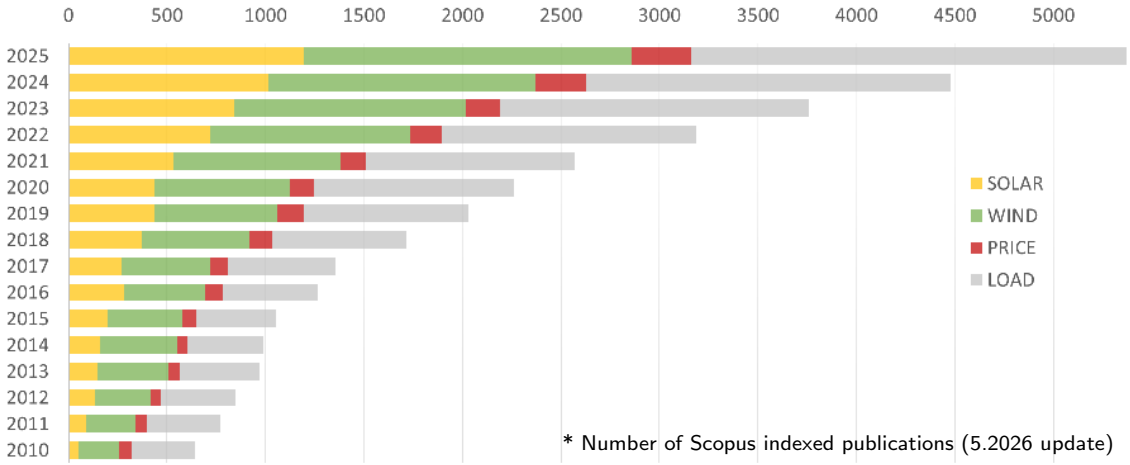


... but the (Academic) Empire



Energy (load, price, wind & solar) forecasting*

(Hong, Pinson, Wang, Weron, Yang & Zareipour, 2020, IEEE OAJPE)



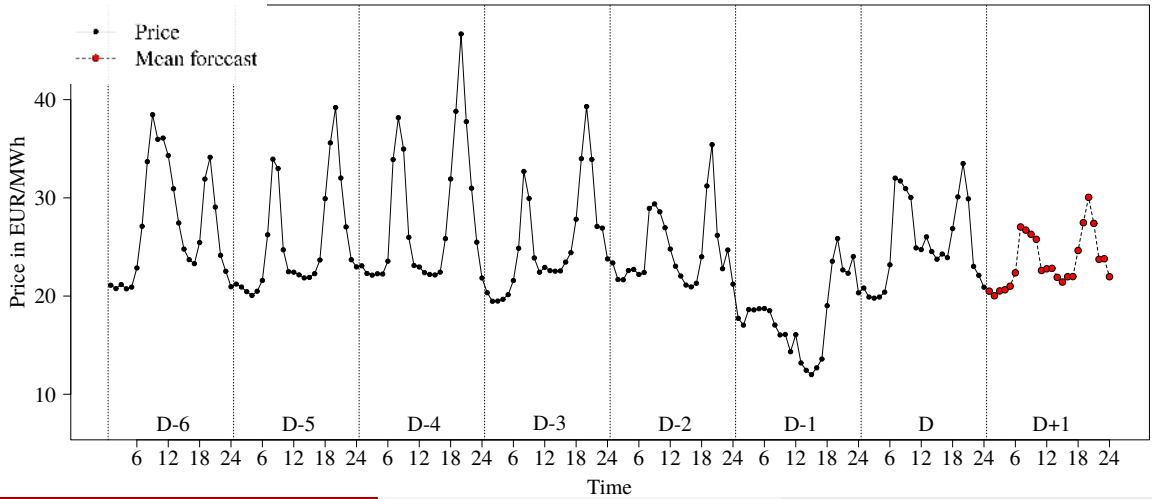
* Number of Scopus indexed publications (5.2026 update)

Day-ahead ($>$ 90% of papers) vs. intraday (real-time) markets

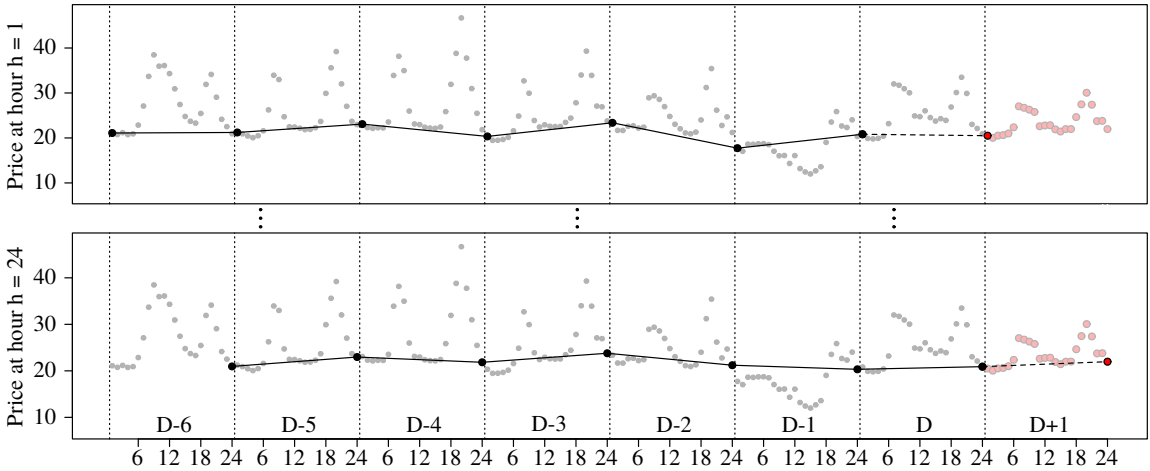
(Maciejowska, Uniejewski & Weron, 2023, Oxford Res. Enc.)

Day-ahead point forecasting: Univariate ...

(Ziel & Weron, 2018, ENEECO)

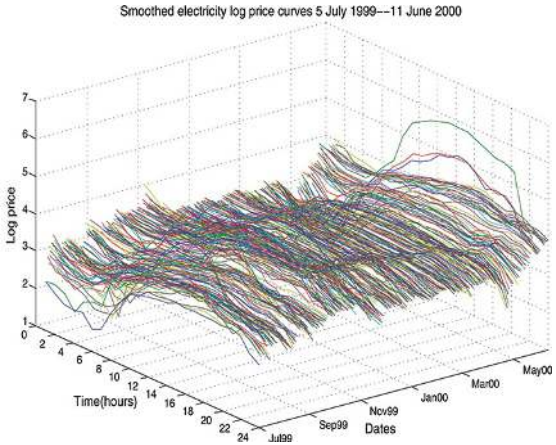


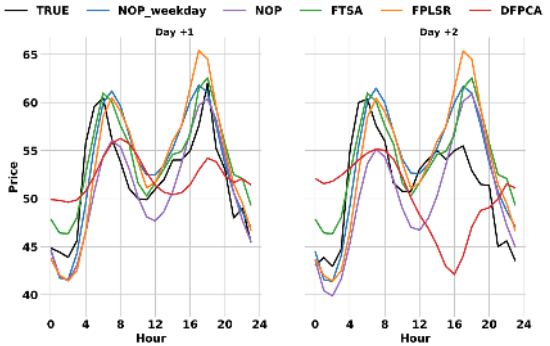
... multivariate ...



... functional (data analysis) ...

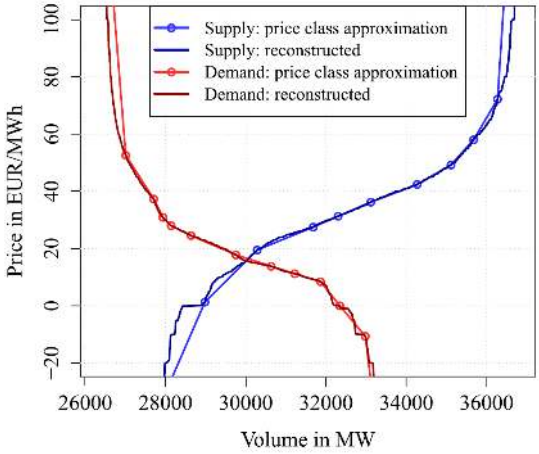
(Chen & Li, 2017, JBES; Chen et al., 2019, Ann.Appl.Stat; Wang & Cao, 2023, Environmetrics)

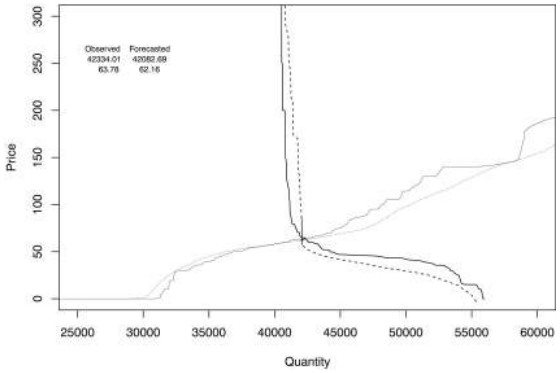




... or supply & demand curves?

(Ziel & Steinert, 2016, ENEECO → 'X-model'; Shah & Lisi, 2020, J.Forecasting)



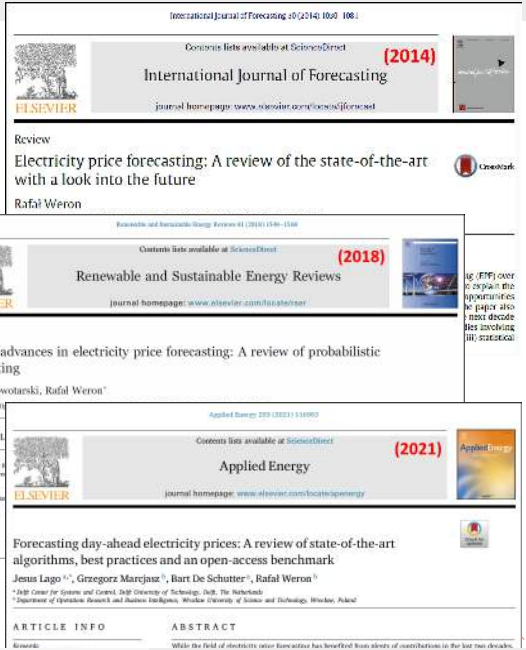


1 Introduction

2 Model evolution
- (Auto)regressive models
- Averaging across windows
- LASSO-Estimated AR (LEAR)
- Deep and interpretable ML

3 Beyond point forecasts

4 Wrap-up



Expert ARX-type models

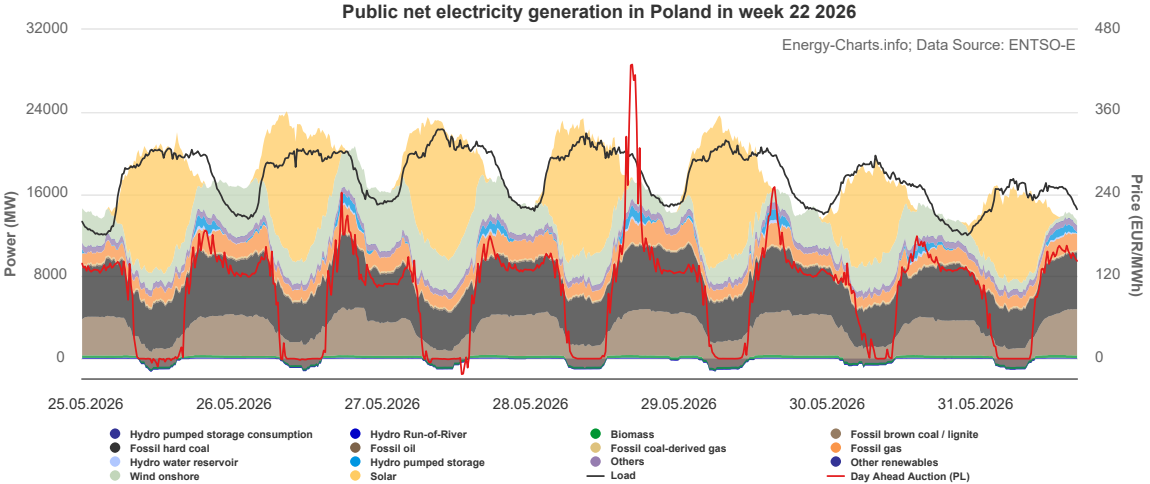
The workhorse of EPF – autoregressive structure with exogenous variables:

$$P_{d,h} = \beta_0 + \underbrace{\sum_{i=1}^{7} \beta_i P_{d-i,h}}_{\text{AutoRegressive effects}} + \underbrace{\sum_{i=1}^{7} \beta_{i+7} D_i}_{D_1 = \text{Mon, ...}} + \underbrace{\sum_{i=1}^{K} \beta_{i+14} X_{d,h}^{(i)}}_{\text{eXogenous variables}} + \varepsilon_{d,h}$$

- Special cases:
  - Naive model $P_{d,h}^{(1)}$ for $\beta_1 = 1$ and $\beta_{i\neq 1} = 0$
  - AR(7), sparse AR(7) with some AR lags missing
  - ARX(7) with $K \geqslant 1$, sparse ARX(7) with some AR lags missing

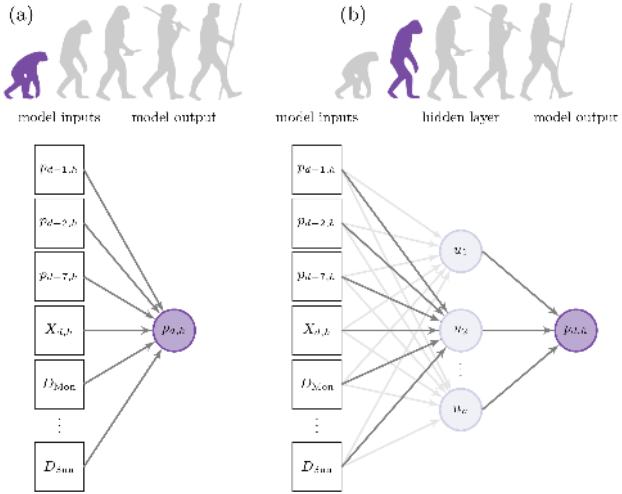


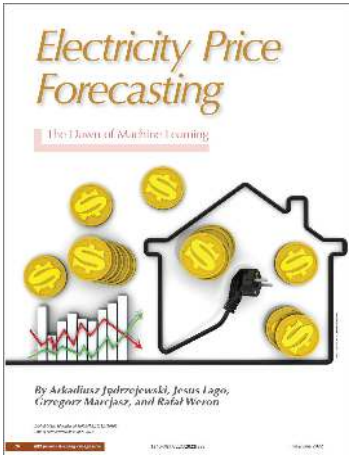
Do we need exogenous variables?



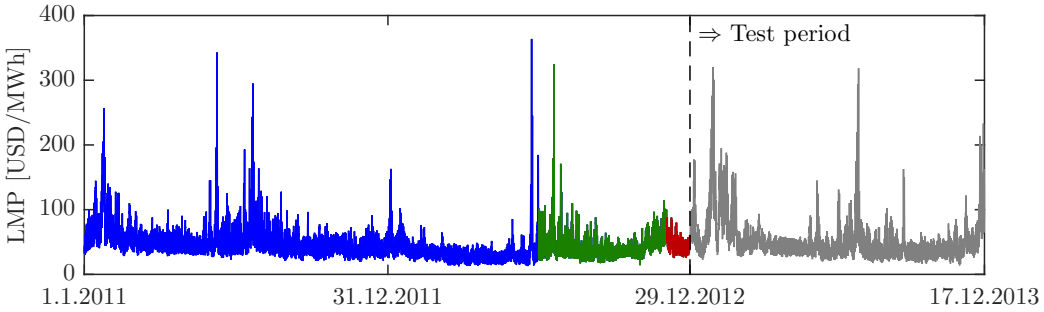
Linear regression vs. single-output (shallow) neural network

(Jędrzejewski, Lago, Marcjasz & Weron, 2022, IEEE-PEM)





What is the optimal calibration (training) window size?



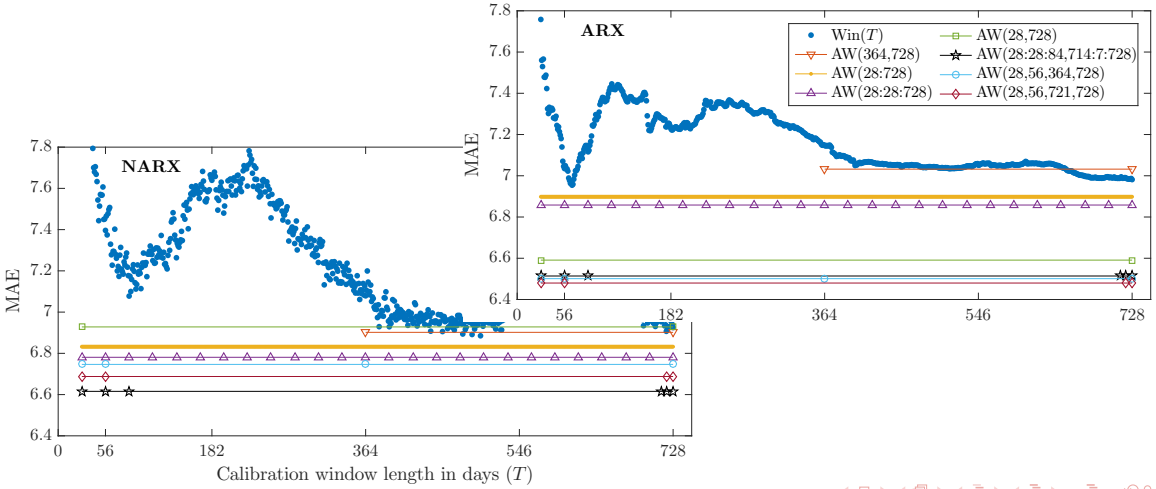
- 728-day vs. 182-day vs. 28-day calibration window
- Longer windows → better reflect trends, more stable parameters
- Shorter windows → quicker adapt to changes in price dynamics

Point forecast averaging: The idea

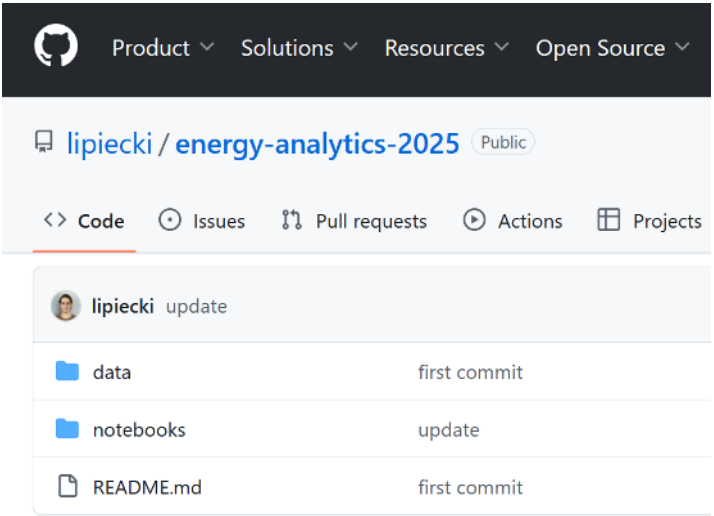
- Dates back to the works of Bates, Crane, Crotty & Granger (1960s)
- 'Statistical/Econometric world': *combining forecasts*, *forecast averaging* (not *model averaging*)
- 'AI/Engineering world': *committee machines*, *ensemble averaging*, *expert aggregation*

Simple averaging across calibration windows

Structural breaks: Pesaran, Pick (2011, JBES); EPF: Hubicka et al. (2019, IEEE-TSE)



IIF Distinguished Lectures on EPF: Averaging.ipynb



Experiment yourself:

What is shrinkage (regularization)?

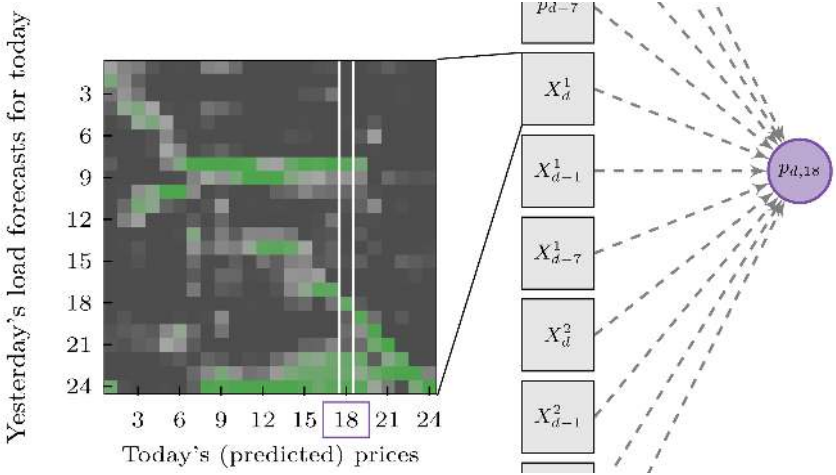
- Minimize the residual sum of squares (RSS $\equiv$ sum of squared errors) + a penalty function of the betas:

$$\hat{\boldsymbol{\beta}} = \underset{\beta_j}{\operatorname{argmin}} \left\{ \underbrace{\sum_{i=1}^{N} \left(y_i - \sum_{j=1}^{p} \beta_j x_{i,j} \right)^2}_{RSS} + \underbrace{\lambda \sum_{j=1}^{n} |\beta_j|^q}_{penalty} \right\}$$

  where $\lambda \geqslant 0$ is a tuning (or regularization) parameter
- Most popular:
  - $q = 2 \rightarrow$ Ridge regression (Hoerl & Kennard, 1970, Technometrics)
  - $q = 1 \rightarrow$ Least absolute shrinkage & selection operator (Tibshirani, 1996, JRSSB)

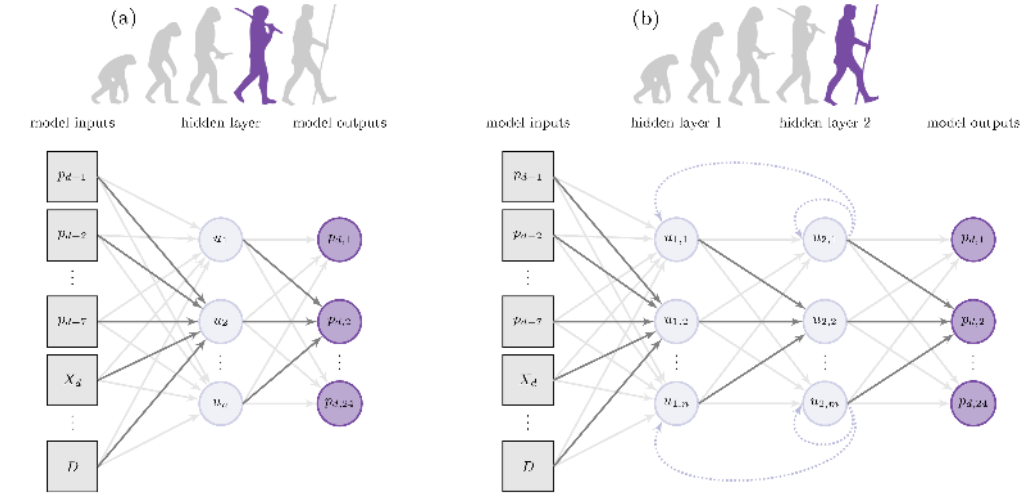
LASSO-Estimated AR (LEAR): Variable importance

(Uniejewski et al., 2016; Ziel, 2016; Ziel & Weron, 2018; Jędrzejewski et al., 2022; Uniejewski, 2024)



Multi-output shallow and deep neural networks (DNNs)

(Lago et al., 2021, APEN; Jędrzejewski et al., 2022, IEEE-PEM)



What about performance?

Contents lists available at ScienceDirect

Applied Energy

(2021)

journal homepage: www.elsevier.com/locate/apenergy

Forecasting day-ahead electricity prices: A review of state-of-the-art algorithms, best practices and an open-access benchmark

Jesus Lago [a,*], Grzegorz Marcjasz [b], Bart De Schutter [a], Rafał Weron [b]

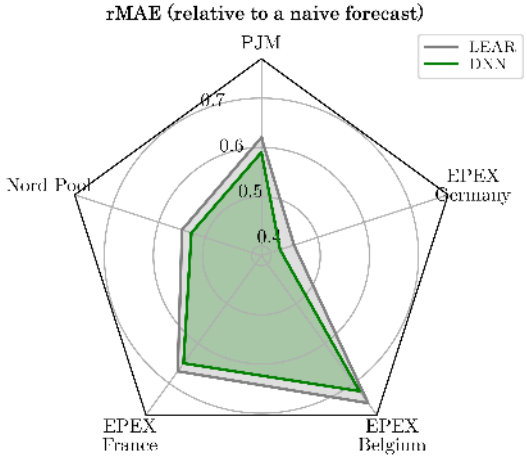
[a] Delft Center for Systems and Control, Delft University of Technology, Delft, The Netherlands
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ARTICLE INFO

Keywords:
Electricity price forecasting
Regression model
Deep learning
Open-access benchmark
Forecast evaluation
Best practices

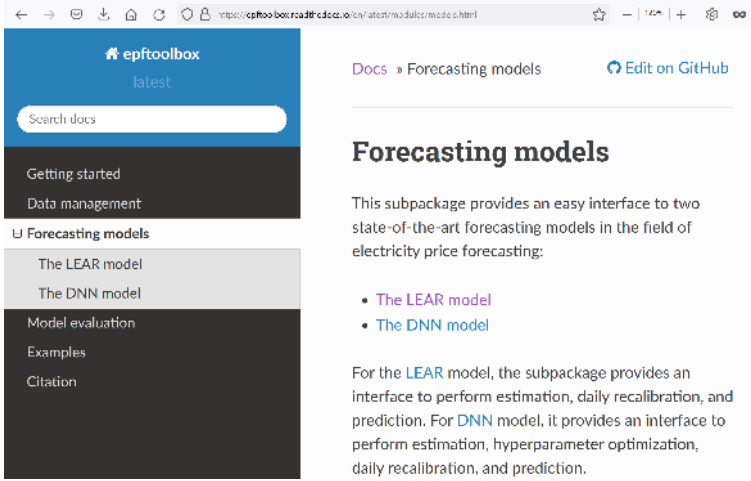
ABSTRACT

While the field of electricity price forecasting has benefited from plenty of c
it arguably lacks a rigorous approach to evaluating new predictive algorit
using unique, not publicly available datasets and across too short and l
The proposed new methods are rarely benchmarked against well estab
models, the accuracy metrics are sometimes inadequate and testing the sig
performance is seldom conducted. Consequently, it is not clear which met
best practices when forecasting electricity prices. In this paper, we tack
of-the-art statistical and deep learning methods across multiple years and
set of best practices. In addition, we make available the considered data
models, and a specifically designed python toolbox, so that new algorit
future studies.



epftoolbox: LEAR and DNN Python codes

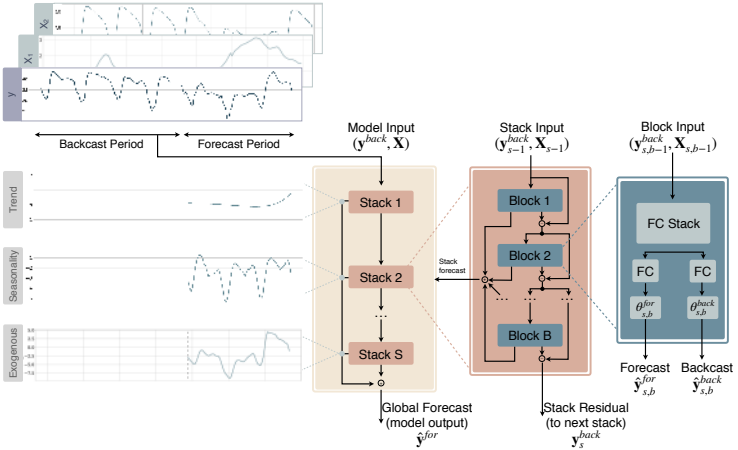
(Lago, Marcjasz, De Schutter & Weron, 2021, APEN)



Experiment yourself:

Interpretable AI: NBEATSx

(Olivares, Challu, Marcjasz, Weron & Dubrawski, 2023, IJF)



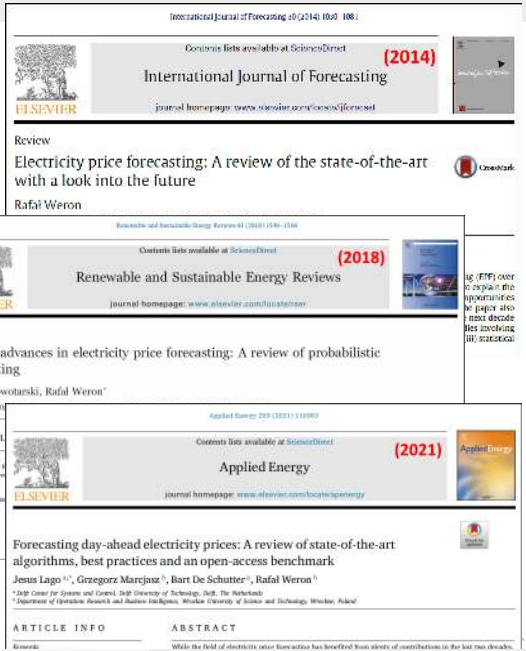
Experiment yourself:

1 Introduction

2 Model evolution

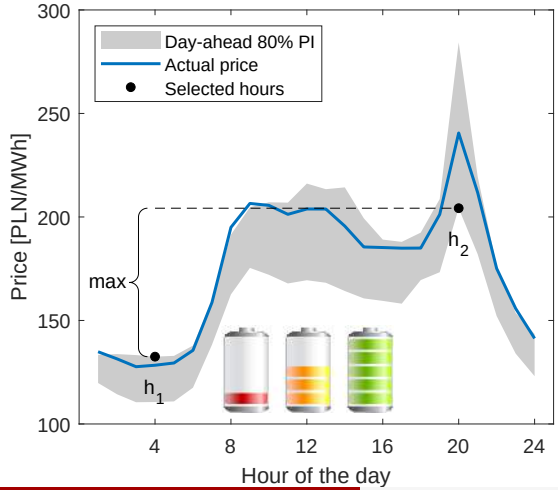
3 Beyond point forecasts
- Do we need probabilistic forecasts?
- Postprocessing point forecasts
- Pinball score and the CRPS
- Distributional Deep Neural Nets
- Transformers

4 Wrap-up



Do we need probabilistic forecasts?

(Uniejewski & Weron, 2021, ENEECO)



Quantile-based trading strategy

Simultaneously bid day-ahead:

- buy 1 MW for $\hat{P}^{1-\alpha}_{d,h_1}$ at hour h_1
- sell 0.8 MW at $\hat{P}^{\alpha}_{d,h_2}$ at hour h_2

$\leftarrow$ With $\alpha = 10\%$ the spread is maximized for $\hat{P}^{90\%}_{d,4} = 133$ and $\hat{P}^{10\%}_{d,20} = 204$ PLN/MWh

- If bid(s) not accepted, settle in the balancing market

Are quantile-based trading strategies more profitable?

(Uniejewski & Weron, 2021, ENEECO)

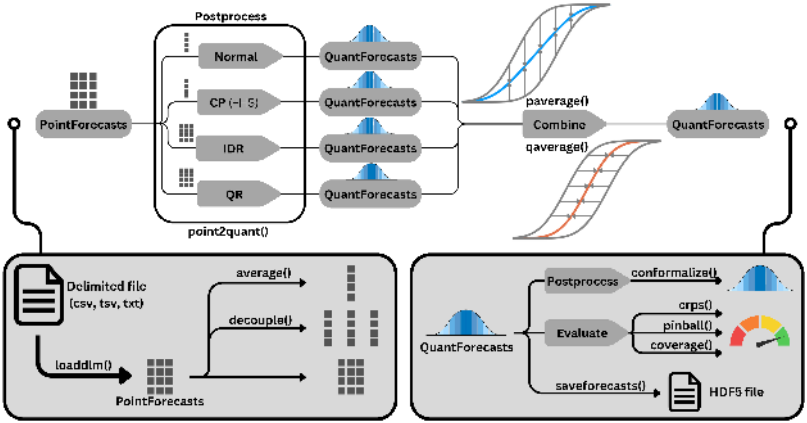


Strategy	Profit				
Naive (4am–12pm)			33 065.29		
Point forecasts-based			33 722.39		
Quantile-based	1-99%	5-95%	10-90%	20-80%	25-75%
Q-Ave	41 317.92	43 328.89	43 432.31	43 289.09	43 033.88
F-Ave	39 848.26	43 369.44	44 052.04	44 088.11	43 130.34
QRM	41 163.29	43 054.28	43 124.12	43 731.54	42 240.25
LQRA(77)	42 360.05	44 135.49	44 713.52	44 684.40	43 624.65
LQRA(BIC)	42 886.10	43 993.23	44 502.81	45 396.21	42 741.57
LQRA(CV)	41 693.80	43 971.88	44 238.45	45 073.19	43 103.88

Quantile-based trading yields 19-34% higher profits than point forecasts-based!

Postprocessing point forecasts with PostForecasts.jl

(Vannitsem et al., 2018, Elsevier; Lipiecki et al., 2024, ENEECO; Lipiecki & Weron, 2025, SoftwareX)



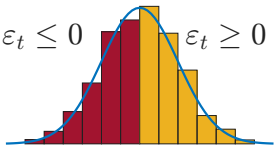
Experiment yourself:

Quantile forecasting

(Lipiecki et al., 2024, ENEECO; Lipiecki & Weron, 2025, SoftwareX)

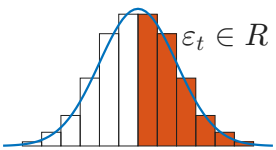
Historical simulation (**HS**)

- Empirical CDF of the prediction errors $\varepsilon_t = y_t - \hat{y}_t$



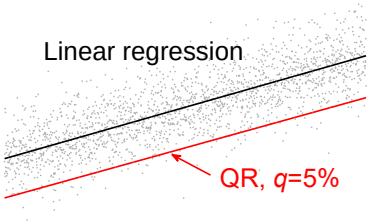
Conformal prediction (**CP**)

- Empirical CDF of $|\varepsilon_t| = |y_t - \hat{y}_t|$
- Smoother but symmetric $\hat{F}$



Quantile regression (**QR**)

- Minimizes the pinball score $\mathrm{PS}(\hat{q}, x, q) = (\mathbb{1}_{\{x<\hat{q}\}} - q)(\hat{q} - x)$



Evaluating reliability & sharpness

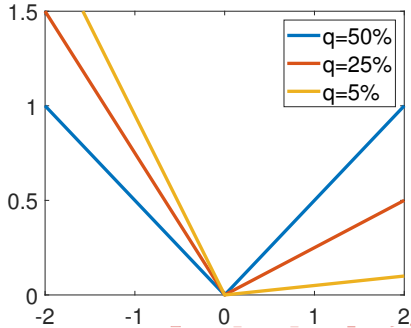
(Gneiting & Raftery, 2007, JASA; Nowotarski & Weron, 2018, RSER; Nitka & Weron, 2023, ORD)

- Pinball score (also linlin/bilinear/newsboy/quantile loss/score, check function):

$$\mathrm{PS}(\hat{q}, x, q) = \left(\mathbb{1}_{\{x<\hat{q}\}} - q\right)(\hat{q} - x) = \begin{cases} (1-q)(\hat{q}-x) & \text{for } x < \hat{q} \\ q(x-\hat{q}) & \text{for } x \geqslant \hat{q} \end{cases}$$

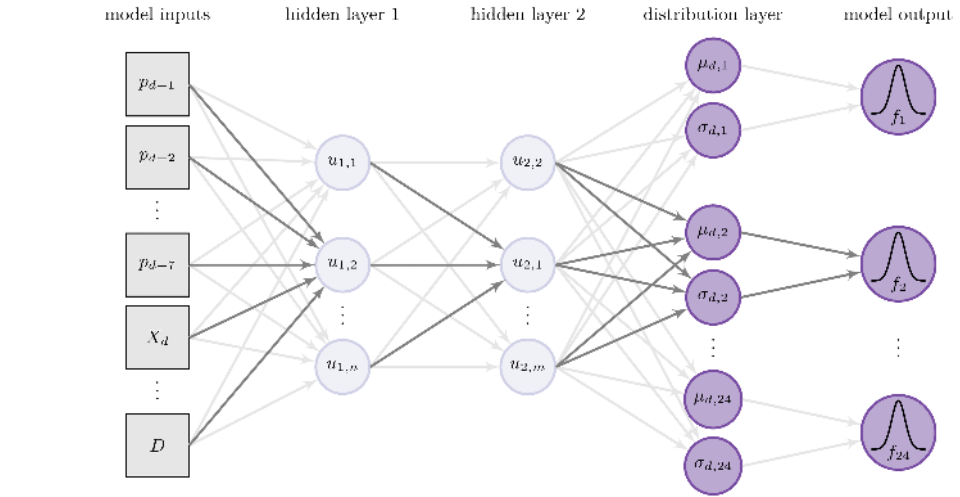
where $\hat{q} \equiv \hat{F}^{-1}(q)$ is the quantile forecast

- For an Aggregate PS (or APS) average:
  - over t in the test period
  - over $q_1 < \ldots < q_M$ quantiles $\rightarrow$ CRPS



Distributional Deep Neural Nets (DDNN)

(Jędrzejewski et al., 2022, IEEE-PEM; Marcjasz, Narajewski, Weron & Ziel, 2023, ENEECO)



DDNNs can perform very well ...

(Marcjasz, Narajewski, Weron & Ziel, 2023, ENEECO)



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Distributional neural networks for electricity price forecasting

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JEL classification:
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Keywords:
Distributional neural network
Probabilistic forecasting

ABSTRACT

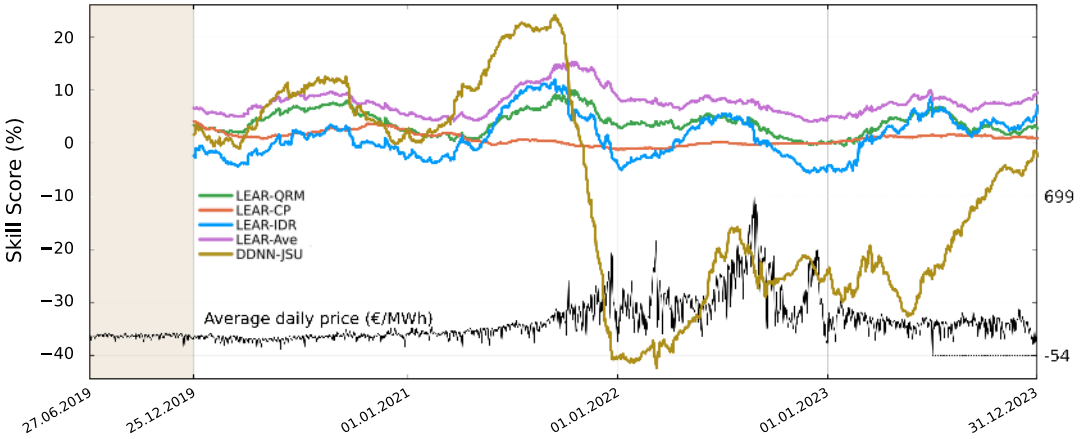
We present a novel approach to probabilistic electricity price forecasting, which utilizes distributional neural networks. The model structure is based on a deep neural network containing a so called probability layer, i.e., the outputs of the network are parameters of the normal or Johnson's SU distribution. To validate our approach, we conduct a comprehensive forecasting study complemented by a realistic trading simulation with day-ahead electricity prices in the German market. The proposed distributional deep neural network outperforms state-of-the-art benchmarks by over 7% in terms of the continuous ranked probability score and by 8% in terms of the per-transaction profits. The obtained results not only emphasize the importance of higher moments when modeling volatile electricity prices, but also – given that probabilistic forecasting is the essence of risk management – provide important implications for managing portfolios in the power sector.

Experiment yourself:

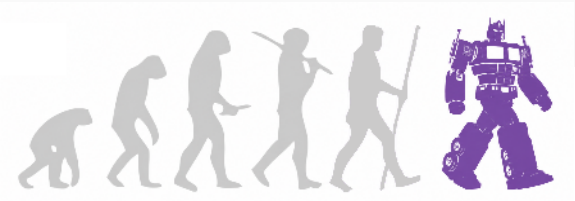
... but can fail during volatile periods → hyperparameters

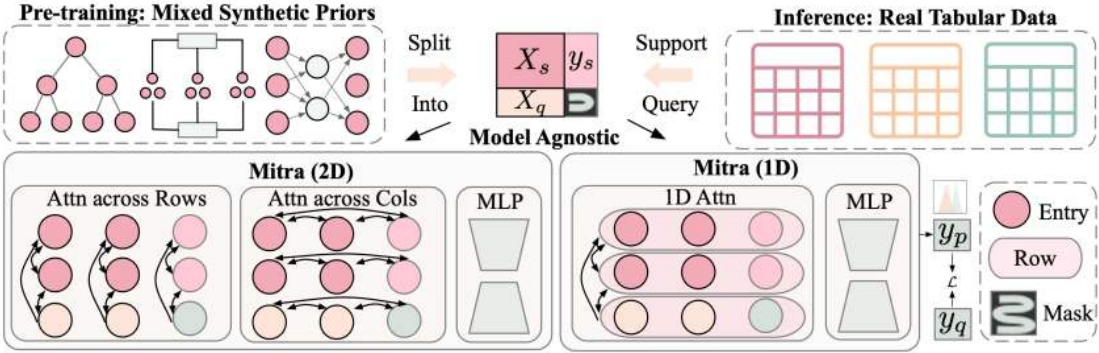
(Lipiecki, Uniejewski & Weron, 2024, ENEECO; illustrated here using EPEX-DE data)

Skill score: % difference wrt the CRPS of N(0,$\hat{\sigma}(\varepsilon_{d,h})$) innovations; $\varepsilon_{d,h}$ – prediction errors of the LEAR model



Chronos, Mitra, TabPFN, ...





Benchmarking "zero-shot" versions against LEAR and DDNN

- Training: 4 year rolling window ($3+1$ for LEAR and Mitra)
- Test: 5 year window (2021-2025), hourly day-ahead prices
- Output: 24 hourly day-ahead prices $\longrightarrow$ expected value + 99 percentiles
- Inputs (features, regressors):

$$p_d = f\Big(\overbrace{p_{d-1},\ p_{d-2},\ p_{d-3},\ p_{d-7},}^{\text{Day-ahead prices}}\ \overbrace{\hat{L}_d,\ \hat{L}_{d-1},\ \hat{L}_{d-7},}^{\text{Load forecasts}}\ \overbrace{\hat{R}_d,\ \hat{R}_{d-1},\ \hat{R}_{d-7},}^{\text{RES forecasts}}$$
$$\underbrace{\text{EUA}_{d-2},\ \text{NG}_{d-2},\ \text{Brent}_{d-2},\ \text{API2}_{d-2}}_{\text{Fundamentals: CO2, gas, crude oil, coal prices}} + \underbrace{D_d}_{\text{Daily dummies}} \Big)$$

Results: RMSE (MAE results similar)

Coloring independent for each year (column)



	2021	2022	2023	2024	2025
LEAR + CP	22.23	45.71	20.75	28.35	20.38
DDNN-Gaussian	27.72	36.60	19.07	25.96	19.50
DDNN-JSU	21.50	37.35	19.00	26.01	20.23
Chronos 2	23.02	41.07	19.43	25.47	20.58
Chronos 2 synth	21.72	41.52	19.59	26.46	20.92
Chronos 2 small	22.76	44.70	21.41	26.62	20.85
TabPFN 2.0	48.32	68.85	18.01	23.40	17.52
TabPFN 3.0	18.51	35.33	17.18	25.02	17.56
TabPFN 3.0 ts	18.03	34.39	17.00	24.53	17.66
Mitra + CP	20.74	38.98	20.18	24.70	19.21

Results: CRPS (Pinball over 99 percentiles)

Coloring independent for each year (column)



	2021	2022	2023	2024	2025
LEAR + CP	5.39	12.59	5.72	5.86	4.71
DDNN-Gaussian	5.47	9.69	4.78	4.85	4.49
DDNN-JSU	4.56	9.86	4.75	4.65	4.56
Chronos 2	4.76	10.81	4.76	4.69	4.55
Chronos 2 synth	4.59	10.73	4.66	4.66	4.51
Chronos 2 small	4.78	11.67	5.41	5.08	4.89
TabPFN 2.0	6.02	11.15	4.15	3.99	3.58
TabPFN 3.0	3.80	9.14	3.99	3.94	3.56
TabPFN 3.0 ts	3.79	8.86	3.93	3.94	3.62
Mitra + CP	4.83	10.72	5.40	4.86	4.55

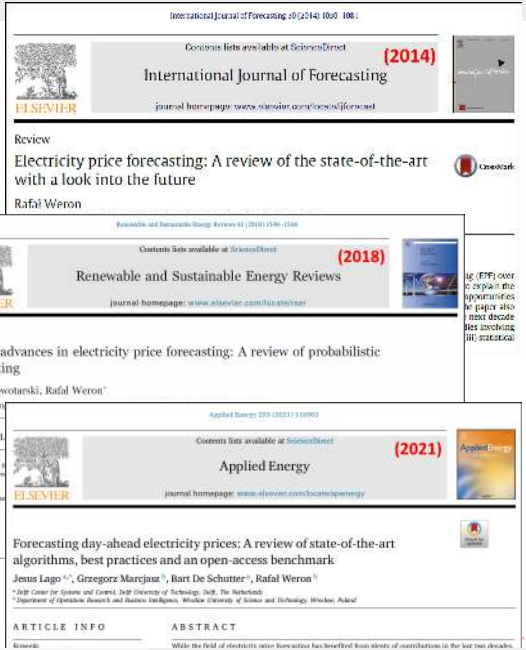
What about profits from BESS trading?

Max 1 BESS cycle per day (1 MWh); coloring across all risk appetites



	90%	80%	70%	60%	50%
LEAR + CP	96 849	104 136	102 979	95 199	85 050
DDNN-Gaussian	106 049	116 433	115 369	108 762	97 965
DDNN-JSU	120 341	123 029	116 784	107 450	97 632
Chronos 2	110 033	121 890	116 714	109 740	99 675
Chronos 2 synth	116 082	121 000	113 343	105 622	96 353
Chronos 2 small	106 372	112 634	105 301	92 055	82 776
TabPFN 2.0	113 002	105 158	94 114	86 199	75 434
TabPFN 3.0	126 422	113 185	101 008	89 587	80 287
TabPFN 3.0 ts	129 230	121 445	111 204	101 222	89 273
Mitra + CP	104 228	100 828	91 725	82 993	73 793

1. Introduction
   - Model taxonomy
2. Model evolution
   - (Auto)regressive models
   - Averaging across windows
   - LASSO-Estimated AR (LEAR)
   - Deep and interpretable ML
3. Beyond point forecasts
   - Do we need probabilistic forecasts?
   - Postprocessing point forecasts
   - Pinball score and the CRPS
   - Distributional Deep Neural Nets
   - Transformers
4. Wrap-up



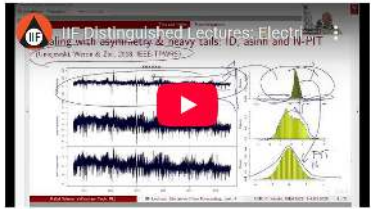
What to learn more? See the IIF Distinguished Lectures

https://p.wz.pwr.edu.pl/~weron.rafal/Conf/IIF24

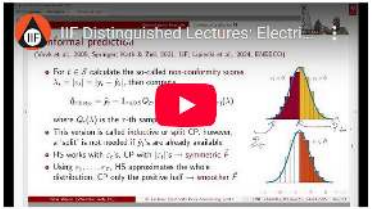
Part I, Lecture I



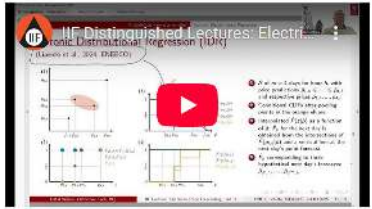
Part II, Lecture I



Part I, Lecture II



Part II, Lecture II



Slides, snippets, movies: